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FADO: A Reaction Controller for Fairness-Aware Online Learning Under Concept Drift

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Abstract:

Online learning systems deployed in non-stationary environments must remain both predictive and fair as the data distribution evolves. Existing fairness-aware online learners optimise a fixed regulariser and do not adapt to concept drift, whereas drift-adaptive streaming methods react to drift but enforce no fairness constraints. We propose Fairness-Aware Drift-Aware Online Learning (FADO), a framework that couples the ARANYANI fair oblique decision forest [1] with a dual-threshold ADWIN drift detector and a reaction controller that modulates the optimiser learning rate and the soft-routing temperature in response to detected drift, and anneals both back to baseline during recovery. We evaluate FADO under a prequential test-then-train protocol on two complementary fairness benchmarks: FOLKTABLES ACS Income (CA), which provides a natural cross-year shift between 2015 training and 2017–2018 prequential test; and a controlled-drift testbed on COMPAS in which a symmetric race swap is paired with a Bernoulli(0.5) label flip to produce an abrupt subpopulation shift. We compare FADO against the ARF and RFR streaming baselines and against a controller-free ARANYANI ablation that isolates the contribution of the drift-reaction controller. Across both datasets and aggregated over 30 random seeds, FADO outperforms ARF and RFR on every reported metric (paired Wilcoxon Holm-corrected $p < 0.001$ throughout), with demographic-parity reductions of ~56%/~56% on FOLKTABLES and ~24%/~25% on COMPAS. The FADO-vs-ARANYANI comparison is regime-dependent: the reaction controller adds a measurable demographic-parity and equalised-odds improvement on the abrupt-shift testbed (COMPAS, both $p < 0.001$) at a sub-0.2pp accuracy trade-off, and is statistically tied with the controller-free baseline on the gradual-shift natural data (FOLKTABLES, sub-0.3pp differences in either direction).

Keywords:

fairness-aware online learning
concept drift
drift detection
group fairness
oblique decision forests

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●Yes ○ No ○ NA	1.1 A clear description of the mathematical setting, algorithm, and/or model.
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